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Career objective

I'm a software engineer with 6 years of experience in software development, results-oriented, target specific with can-do attitude, actively involved in the open-source community, looking for an opportunity that provides stability, career development where I can learn new technologies and add value. I'm capable of working individually or as a part of a team, well organized and trustworthy.

Education

- **2013 - 2017**: B.Sc. in engineering (electronics and communications department), Cairo university.
  - **Grade** :73% (Good).
  - **Graduation project**: Decentralized wireless network node (DWNN) prototype, grade: (Excellent).
- **2011 - 2012**: Secondary school, grade: 99.9%.

Personal experience

- **Dec 2025 - present:** Creating LLG-MC "Language Learning Gamification Media Center" on GitHub (a comprehensive multi-modal media center application with a three-tier Client-Server-Dashboard architecture integrating video/audio playback, book reading and web content management):
  - Media player with clickable/translatable subtitles that can be saved for future reference.
  - Book reader with bookshelf and OCR (clickable/translatable/saved) text overlay.
  - Real-time communication via websocket for shared viewing/reading/chatting sessions**.**
  - Administrative dashboard with push notifications, license/users management, incremental updates.
  - Backend server with authentication/authorization scheme using JWT tokens & OAuth2 integration.

- **Apr 2026 – present**: JARVIS Personal Assistant (AI Voice Assistant with speech recognition & Computer Vision):
  - Designed modular architecture with 7 integrated components (Brain, Ears, Mouth, Actuators for hands/gesture tracking, Eye, Memory, Sound Manager) using Python FastAPI microservices.
  - Implemented AI/ML capabilities using PyTorch including LoRA/Unsloth for LLM fine-tuning.
  - Integrated vector database for long-term memory storage and context-aware conversation history.
  - Built real-time speech-to-text (Faster-Whisper), and text-to-speech (Kokoro) with JavaFX frontend.

- **Jan 2026 – present: Invent OR Invest** (MERN-Stack collaboration platform connecting inventors & investors together):
  - Built comprehensive project management system with role-based team collaboration & reporting**.**
  - Developed AI-powered identity verification using face-api.js for secure user authentication.
  - Created commitment letter generation system with PDF export functionality
  - Built reputation system with ratings, reviews, and trust scoring algorithms
  - Designed advanced search and filtering with pagination, sorting, and category-based discovery
  - Implemented document management system for business plans, technical specs, and legal documents

- **Apr 2019 – Jun 2021**: Creating LLG-MP "Language Learning Gamification Media Player" on GitHub.

Trainings & Courses

- **Jan 2020 - Jun 2020:** AWS cloud DevOps nanodegree **by** "Udacity".
- **Mar 2020 - Aug 2020:** AWS cloud developer nanodegree **by** "Udacity".
- **Jun 2019 - Nov 2019:** Deep Learning nanodegree **by** "Facebook AI team on Udacity - NTL scholarship".
- **2018:** Cisco AppDynamics onsite training **by** "Cisco Systems in Dubai".
- **2018:** Splunk certified user & Splunk certified power user **by** "Splunk corporation".

Development skills

- **Programming languages:**
  - Java [Experienced]
  - JavaScript [Experienced]
  - SQL [Experienced]
  - Python [Used]
  - C/C++/C# [Used]
  - NoSQL [Used]
- **Frameworks:**
  - Spring Boot [Experienced]
  - Node.js [Experienced]
  - JPA/Hibernate/JUnit [Experienced]
  - Angular.js [Used]
  - HTML/CSS/Bootstrap [Used]
  - Ajax/jQuery [Used]
- **Architecture:**
  - OOP/Data structures/Algorithms
  - GraphQL
  - Microservices
  - Design patterns
  - ORM/TDD
  - Big data
  - Rest/Soap APIs
  - CI/CD
  - Map reduce
- **Tools:**
  - VSCode/Eclipse/STS/IntelliJ
  - Spark/Hadoop/Kafka
  - Elastic search
  - JMeter
  - Redis/Memcached
  - Docker/Kubernetes
  - Postman/SoapUI
  - RabbitMQ/JMS
  - Git/Jenkins
- **Languages:**
  - Arabic (Native)
  - English (Fluent)
- **References:**
  - Portfolio
  - Github
  - Stackoverflow
- **Books:**
  - Clean Code
  - Clean Coder
  - Clean Architecture
  - Clean Craftsmanship
  - The Pragmatic programmer
  - Code Complete

Work experience:

- **Jul 2020 – June 2021:** Senior software application engineer at Orange:
  - Customizing big data solutions according to the customer's needs (BMW account).
  - Development of custom plugins ensuring adherence to automated business workflows.
  - Analyzing business pains and customer requirements to ensure delivery of data intensive applications.

- **Jan 2019 – Jun 2020:** Software engineer at DG-BI (a software house located in Heliopolis):
  - Delivered **Swift watcher project** (a banking transactions validation switch):
    - Built APIs through which a copy of banking transactions is sent to a central server.
    - Created a listener service that checks transactions against sanction/AML rules stored in SQL database.
    - Designed & implemented a user interface for agents to validate malicious transactions violating swift. rules for further investigation and reporting.
    - Performed unit/load tests for system components to assert prespecified functionalities and performance metrics
  - Delivered **generic configuration parser project**:
    - Designed and implemented a central parsing engine that digests the configuration files of PS department's solutions such as [Fawry Payment – Mobile wallet – visual Journal, ...etc.].
    - Created the user interface with authentication & authorization schemes for the support agents to control configuration parameters.

- Delivered **Fawry service client-side application project**:
  - Created the required wireframe along with various customer modifications.
  - Created the delivery scheme & performance metrics of the product.
  - Implemented the solution development, testing & deployment phases.
  - Documented the project using swagger.
- Participated in the delivery of **central monitoring system project**:
  - Designed & implemented the system data layer (Hibernate).
  - Designed & implemented the front-end UI (using Angular.js) according to the customer needs.

- **Sep 2017 – Dec 2018:** Business solutions system engineer at Barq systems (a solutions integrator located in Maadi):
  - Participated in the implementation of **Cisco AppDynamics APM project** at bank Al-Bilad:
    - Performed product customization, implementation & integration with the core banking and exchange server.
    - Implemented various [role-policy-action] monitoring sets with their associated reports and dashboards.
    - Performed a project scaling POC resulting in justifying the need for the hardware requirements resulting in IT division acceptance of the requested resources & equipment.
  - Designed & implemented the **truck tracking system project** for the storage department:
    - Created an android application that reports the trucks' location & drivers' personal information to a central server on real time basis.
    - Visualized the sent coordinates on the tracker application (Splunk Mint) along with drivers' metadata.
  - Designed & implemented **news feed application project** for Microsoft SharePoint:
    - Created SharePoint web part that grabs the news feed in XML format from Barq systems official website then injects it in an intranet SharePoint blog for employees to like, comment and share its content.
  - Participated in the implementation **health care CRM (Tabibi) project**:
    - Used best practices to customize the solution's backend & front end according to the customer needs.
    - Developed custom modules, views extensions & webhooks for Sugar CRM.
  - Participated in some **discrete operational work**:
    - Increased speed, scale & reliability of the department's CI/CD pipeline by Integrating Jenkins with OpenShift.
    - Improved team collaboration process through implementation of task management system using Jira.
    - Revamped the documentation system of the department by implementing confluence knowledge base.

- **Jun 2017 – Aug 2017: Freelancing an eCommerce web application**:
  - Gathered requirements, designed, developed, tested, containerized, deployed and documented the UAT & production environments.
  - Assembled authentication module (Registration/Verification/Login/Password reset) with CSRF protection.
  - Developed dynamic admin panel for the business owner to (add/delete/modify) products & categories lists, observe & deliver customers purchase requests.
  - Deployed image hosting service Integrated with flicker API through authorization using OAuth protocol.
  - Implemented different user authorization workflow for both customers and sales agents.